

AmropaliNet: Detect early. Spray less. Waste less.

AI mango doctor for the Amrapali farmers of Bangladesh - protecting orchards, soil, water and bees

Problem Statement

272 words

Amrapali is Bangladesh's favourite everyday mango and the main source of income for thousands of farming families in Rajshahi, Chapainawabganj, Naogaon and the hill districts. Every season it is attacked by six common diseases: anthracnose, bacterial canker, powdery mildew, scab, sooty mould and stem-end rot.

Most farmers cannot tell these diseases apart, and an agricultural officer is rarely at hand when the first spots appear. The usual answer is blanket, calendar spraying: the whole orchard is sprayed with several fungicides and insecticides "just in case", again and again from flowering to harvest.

This turns a farming problem into an environmental one. Surplus chemicals wash into soil, ponds and groundwater, harming aquatic life and village drinking water (SDG 6). Spraying at flowering kills the bees and other pollinators that mango trees need to set fruit, and repeated doses breed resistant pathogens (SDG 15). Farm workers and their families breathe and handle these chemicals, often without protection (SDG 3).

The second problem is waste. When disease is noticed late, it has already spread, and infected fruit rots on the tree, in storage or in transport. Every discarded mango wastes the water, land, fertiliser, fuel and labour used to grow it, and rotting fruit releases greenhouse gases (SDG 12, SDG 13). Lost harvests and wasted chemical costs push small farmers towards poverty and shrink the local food supply (SDG 1, SDG 2).

ROOT CAUSE

The root cause is one missing piece of knowledge at the right moment: which disease is this, and what is the smallest treatment that works? Without it, farmers overspray, lose fruit and damage the ecosystem their orchards depend on - season after season.

Environmental harm and the SDG targets it violates

Environmental problem	SDG	Target that is violated
Chemical runoff into soil, ponds and groundwater	SDG 6	6.3 Improve water quality by reducing pollution and hazardous chemicals
Blanket, calendar spraying of agro-chemicals	SDG 12	12.4 Sound management of chemicals; less release to air, water and soil
Pollinators killed, resistant pathogens bred	SDG 15	15.5 Halt the loss of biodiversity
Farm families exposed to pesticides	SDG 3	3.9 Fewer illnesses from hazardous chemicals and pollution
Diseased fruit rots before it is eaten	SDG 12	12.3 Reduce food losses along production and supply chains
Wasted water, land, fuel and emissions	SDG 13	13 Take urgent action to combat climate change and its impacts
Lost harvests and income, less local food	SDG 2	2.4 / 1.5 Sustainable, resilient food production; resilience of the poor

Where: Rajshahi, Chapainawabganj, Naogaon and the hill districts

Who is affected: small mango farmers, farm workers, consumers, bees and orchard wildlife

AmropaliNet: the right treatment, only where needed

One photo, the right diagnosis, less chemical and less waste - a healthier environment

Solution

282 words

AmropaliNet puts an expert mango doctor in every farmer's pocket, so that treatment decisions are based on evidence, not fear.

The farmer takes one photo of a mango with any phone. A CLIP-based check first confirms that the photo really shows a mango, so a wrong picture never leads to wrong advice. Our AA-ENet model (EfficientNet-B0 with CBAM attention and a Transformer encoder), trained on 3,500 real Amrapali images, then identifies the disease - or confirms the fruit is healthy - in seconds. Grad-CAM marks the affected area, so the farmer can see why the AI decided and trust the result.

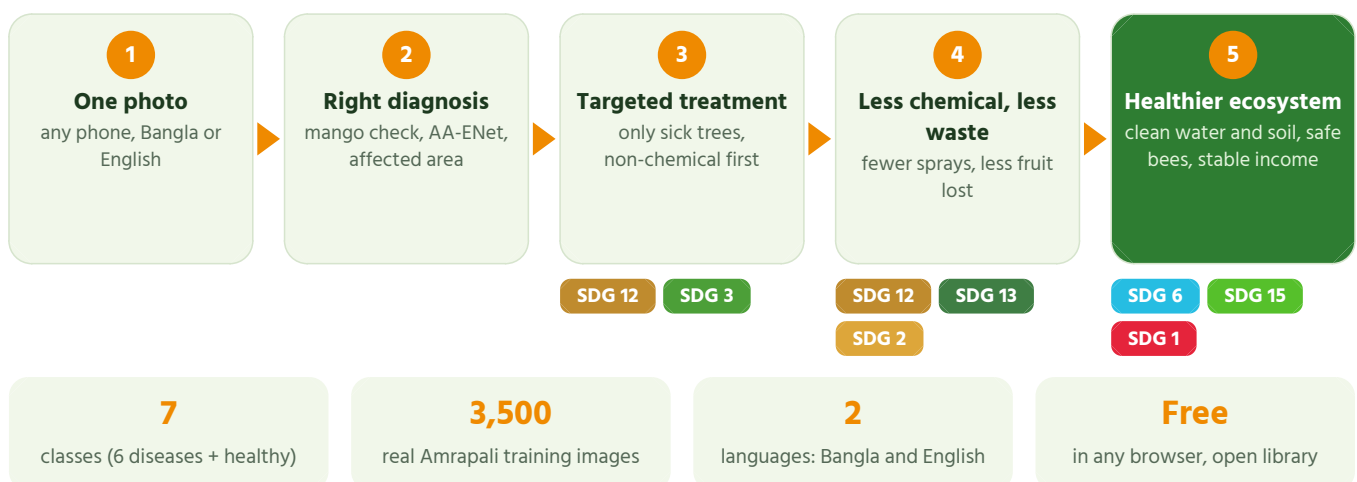
Advice comes in simple Bangla or English: treat only the sick trees, with the right product at the right time, and start with non-chemical steps such as pruning, orchard hygiene, neem oil and hot-water treatment after harvest. A healthy result means no spray at all. A one-page PDF report can be shown to an agricultural officer or buyer.

Each decision is small, but together they change the orchard environment:

- Targeted treatment replaces blanket spraying, cutting the chemical load on soil and water (SDG 6, SDG 12).
- Fewer sprays at flowering protect bees and orchard biodiversity (SDG 15).
- Less exposure protects the health of farming families (SDG 3).
- Early detection stops spread, so less fruit - and the water, land and emissions behind it - is wasted (SDG 12, SDG 13).
- Lower costs and saved harvests mean steadier income and more local food (SDG 1, SDG 2).

AmropaliNet runs free in any browser, works on low-cost phones and is published as the open Python library mango-disease-ai, so agricultural services and other developers can build on it. Detect early. Spray less. Waste less.

How everything connects: from one photo to the SDGs



SDGs AmropaliNet advances:

SDG 1 No Poverty

SDG 2 Zero Hunger

SDG 3 Good Health

SDG 6 Clean Water

SDG 12 Responsible Production

SDG 13 Climate Action

SDG 15 Life on Land